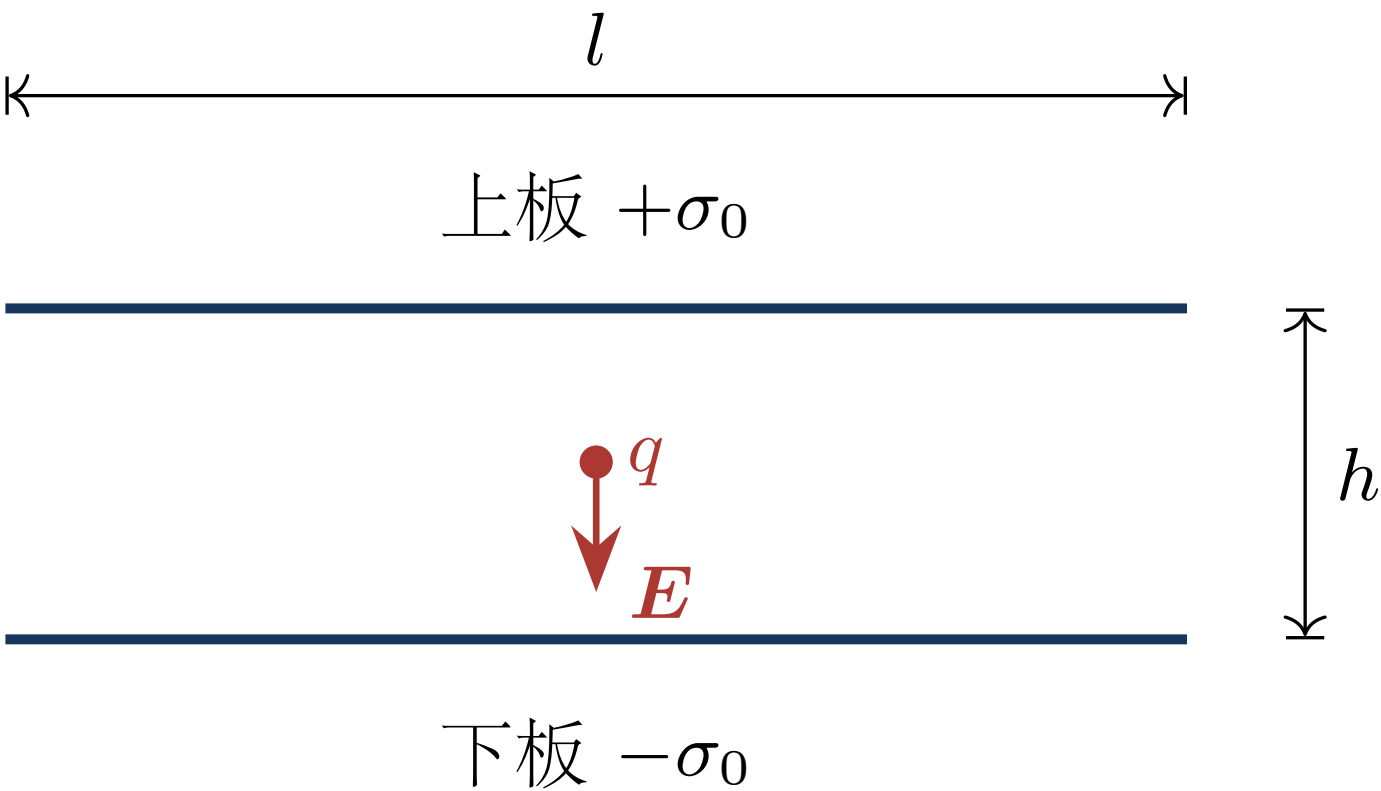


法向截面示意，$h \ll l$

S：电容静止

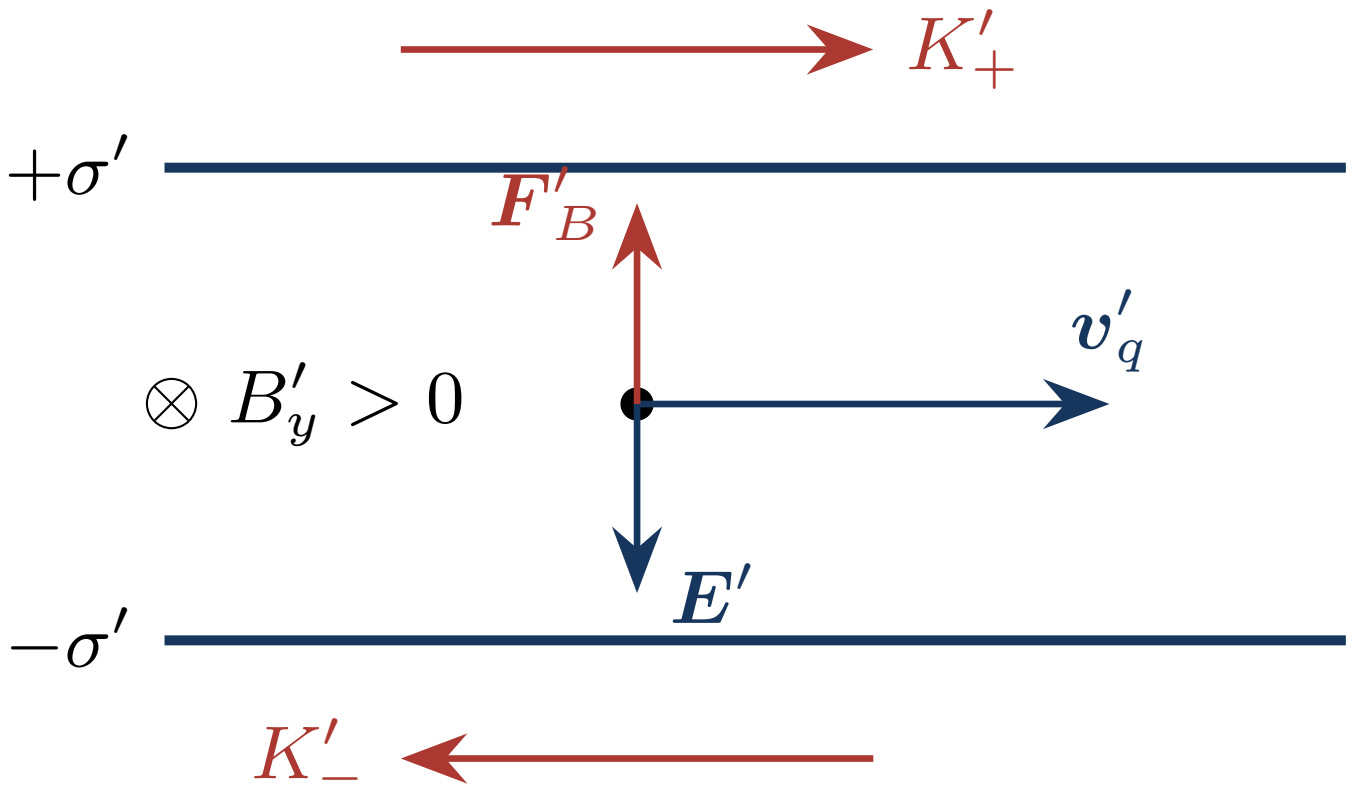
$$A = l^2$$
$$Q = \sigma_0 l^2$$

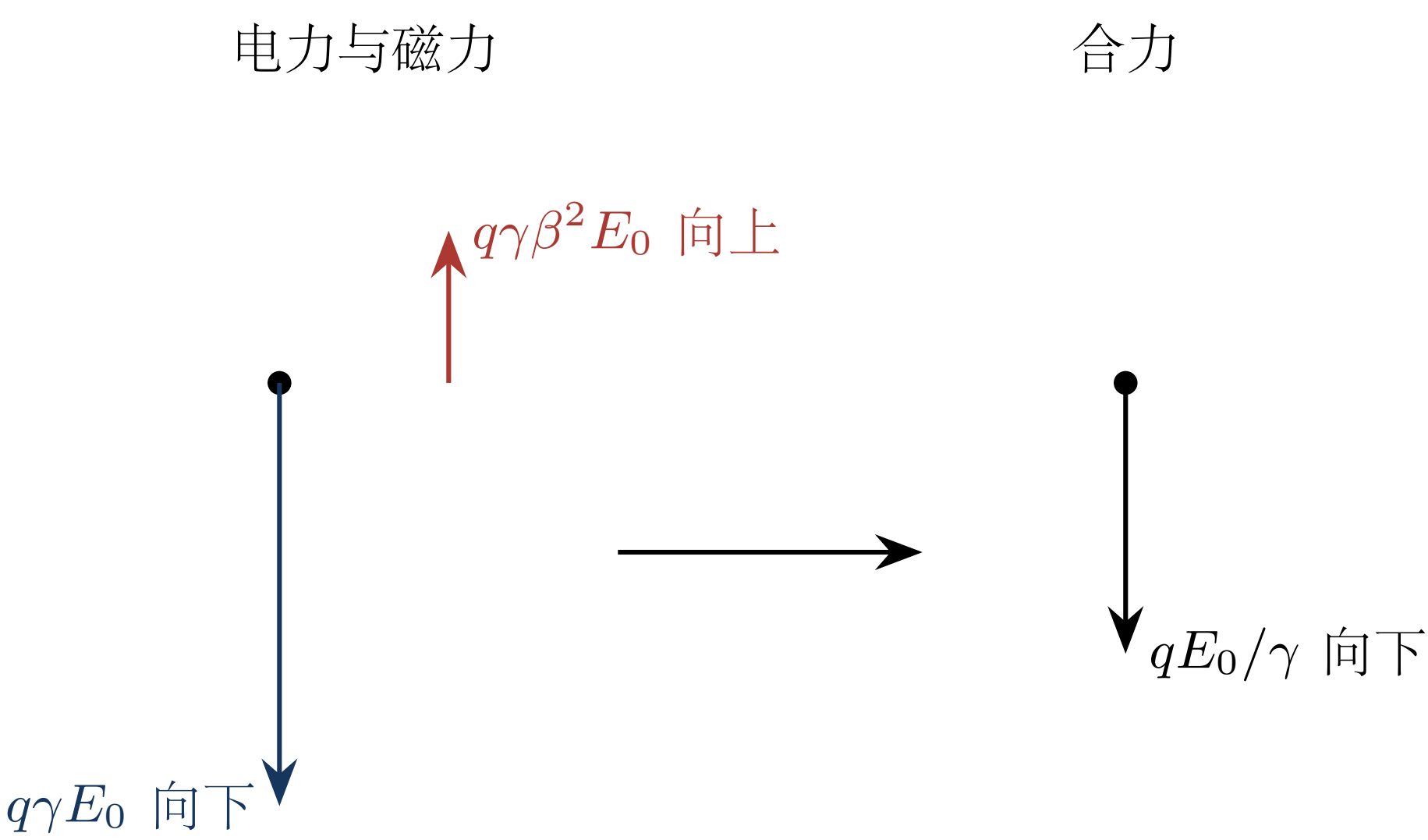
S'：电容向右

$$A' = l^2/\gamma$$
$$\sigma' = \gamma\sigma_0$$

总电荷不变，沿 x 方向长度收缩

$+x$ 向右、$+z$ 向上、$+y$ 入纸面


K'_+
$+\sigma'$
$\boldsymbol{F}'_B$
$\boldsymbol{v}'_q$
$\otimes\ B'_y > 0$
$\boldsymbol{E}'$
$-\sigma'$
K'_-

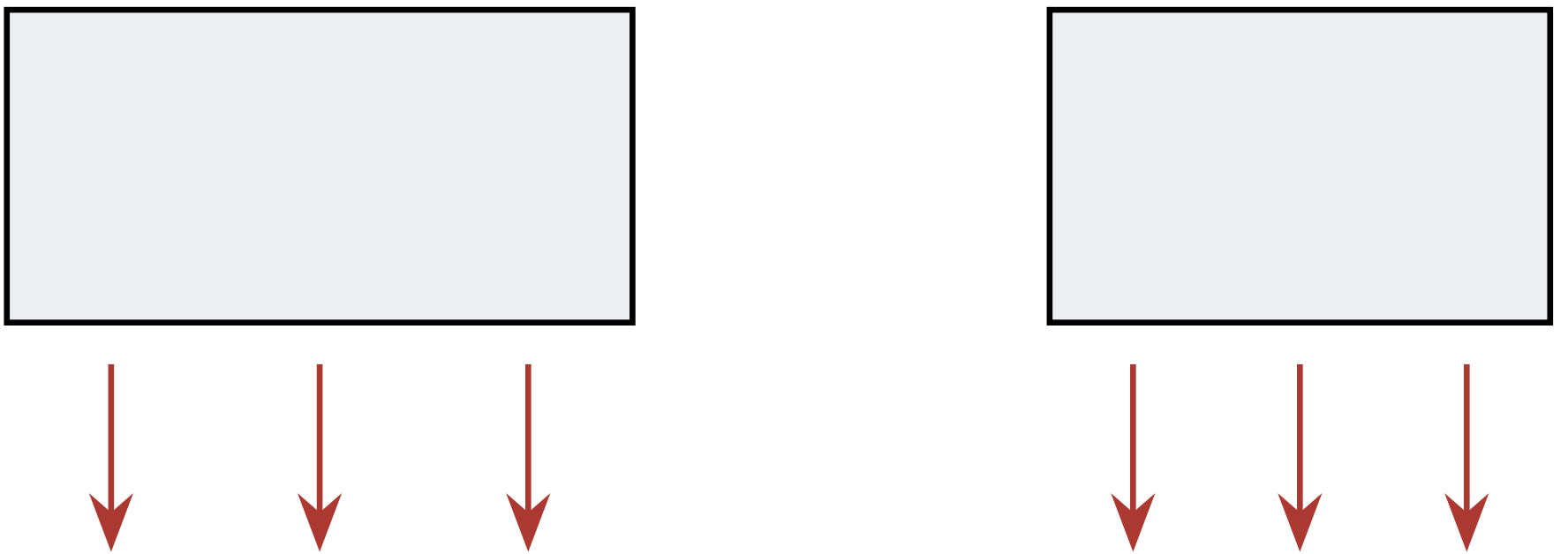


示例 $\beta = 0.6$，$\gamma = 1.25$

四力核对：$K'^z = K^z = \gamma F'_z$

原系面积 l^2

运动系面积 l^2/γ



$p = \sigma_0^2/(2\varepsilon_0)$

$p' = p$

$$F' = p'A' = F/\gamma$$

$$T'_{zz} = \varepsilon_0(E'^2 - c^2 B'^2)/2$$